Batteries, Fuel Cells and Corrosion

Answer key with worked solutions

A. Batteries and fuel cells

1. Batteries.

> (a) A **primary** battery is used once; its reaction cannot practically be reversed. A **secondary** battery can be recharged. (b) An external current forces electrons in the opposite direction, driving the cell reaction **backwards** (electrolysis): Li^+ ions move back into the graphite electrode, restoring the reactants.

2. Fuel cell.

> Anode: $2\,H_2 \longrightarrow 4\,H^+ + 4\,e^-$; cathode: $O_2 + 4\,H^+ + 4\,e^- \longrightarrow 2\,H_2O$; overall: $2\,H_2 + O_2 \longrightarrow 2\,H_2O$
>
> $E^\circ_{\text{cell}} = 1.23\ \text{V} - 0.00\ \text{V} = \mathbf{1.23\ V}$
>
> Advantage: the only exhaust is **water** (and it doesn't run down while fuelled). Challenge: hydrogen is hard to **store** and is often made from fossil fuels.

B. Corrosion

3. Three test tubes.

> Tube 1: nail in tap water open to air: **rusts**. Tube 2: nail in boiled water (oxygen removed) with a layer of oil on top: **no rust** (no oxygen). Tube 3: nail in dry air with a drying agent (anhydrous calcium chloride), stoppered: **no rust** (no water). Only the tube with both water and oxygen shows rust.

4. Rusting half-equations.

> Anode: $Fe \longrightarrow Fe^{2+} + 2\,e^-$; cathode (neutral water): $O_2 + 2\,H_2O + 4\,e^- \longrightarrow 4\,OH^-$
>
> In acid: $E^\circ_{\text{cell}} = 1.23\ \text{V} - (-0.44\ \text{V}) = \mathbf{1.67\ V}$

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5. Coatings.

> (a) Zinc is **more reactive** than iron (E° −0.76 V versus −0.44 V), so where both are exposed, zinc is oxidized instead of iron: **sacrificial protection**. (b) Tin is **less reactive** than iron, so at a scratch the iron becomes the **anode** and corrodes, faster than if it were bare.

C. Calculations

6. AA cell.

> (a) $2.500\ \text{A h} \times 3600\ \text{s/h} = \mathbf{9000\ C}$
>
> (b) $\dfrac{9000\ \text{C}}{96\,485\ \text{C/mol}} = \mathbf{0.0933\ mol\ e^-}$
>
> (c) 2 mol e^- per mol Zn: $\dfrac{0.09328\ \text{mol}}{2} \times 65.38\ \text{g/mol} = \mathbf{3.05\ g\ Zn}$
>
> (d) $9000\ \text{C} \times 1.5\ \text{V} = 1.35 \times 10^4\ \text{J}$, about **14 kJ**

7. Hydrogen used.

> Charge: $1.00\ \text{A} \times 3600\ \text{s} = 3600\ \text{C}$; electrons: $\dfrac{3600\ \text{C}}{96\,485\ \text{C/mol}} = 0.03731\ \text{mol}$
>
> H_2 (2 e^- each): $0.01866\ \text{mol} \times 2.016\ \text{g/mol} = \mathbf{0.0376\ g}$

D. Challenge

8. Sacrificial anode.

> Moles of Mg: $\dfrac{5000\ \text{g}}{24.31\ \text{g/mol}} = 205.7\ \text{mol}$; electrons: $2 \times 205.7 = 411.4\ \text{mol}$
>
> Charge: $411.4\ \text{mol} \times 96\,485\ \text{C/mol} = 3.97 \times 10^7\ \text{C}$
>
> Time: $\dfrac{3.97 \times 10^7\ \text{C}}{0.50\ \text{A}} = 7.94 \times 10^7\ \text{s}$; $\dfrac{7.94 \times 10^7\ \text{s}}{3.156 \times 10^7\ \text{s/year}} = \mathbf{about\ 2.5\ years}$

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